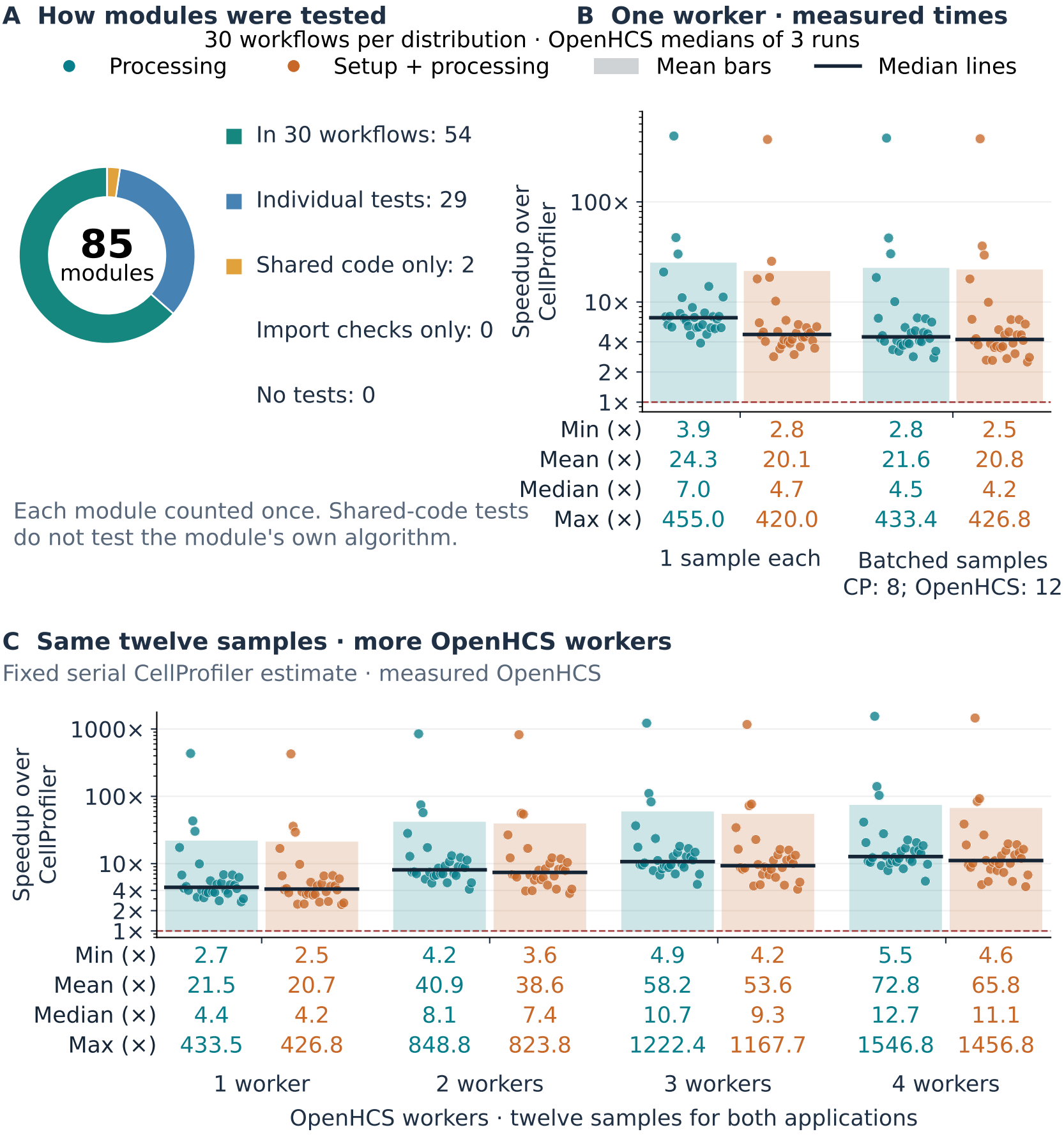
A How modules were tested
B One worker · measured times
30 workflows per distribution · OpenHCS medians of 3 runs
Processing
Setup + processing
Mean bars
Median lines
85
modules
In 30 workflows: 54
Individual tests: 29
Shared code only: 2
Import checks only: 0
No tests: 0
Each module counted once. Shared-code tests do not test the module's own algorithm.
Speedup over CellProfiler
100×
10×
4×
2×
1×
Min (×) 3.9 2.8 2.8 2.5
Mean (×) 24.3 20.1 21.6 20.8
Median (×) 7.0 4.7 4.5 4.2
Max (×) 455.0 420.0 433.4 426.8
1 sample each
Batched samples
CP: 8; OpenHCS: 12
C Same twelve samples · more OpenHCS workers
Fixed serial CellProfiler estimate · measured OpenHCS
Speedup over CellProfiler
1000×
100×
10×
4×
2×
1×
Min (×) 2.7 2.5 4.2 3.6 4.9 4.2 5.5 4.6
Mean (×) 21.5 20.7 40.9 38.6 58.2 53.6 72.8 65.8
Median (×) 4.4 4.2 8.1 7.4 10.7 9.3 12.7 11.1
Max (×) 433.5 426.8 848.8 823.8 1222.4 1167.7 1546.8 1456.8
1 worker
2 workers
3 workers
4 workers
OpenHCS workers · twelve samples for both applications